

IRONHAVEN

SAFETY INTELLIGENCE

Real-Time Underground Tunnel Safety Monitoring

Powered by AI • Built for Lives • 23 Languages

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AI Agent

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THE CRISIS

Teesta Stage-VI Tunnel Disaster — July 2, 2026

A catastrophic tunnel collapse in Sikkim, India trapped **30+ workers** underground. Rescue teams had **no real-time sensor data**, **no digital communication** with trapped crews, and **no centralized command system**.

200+

Active tunnel projects

50–80

Deaths per year

0

Digital monitoring systems

20+

Languages spoken by crews

"Every minute matters underground. The difference between a safe evacuation and a tragedy is just seconds of early warning."

THE SOLUTION

What is Ironhaven?

Ironhaven Safety Intelligence is a **real-time, AI-built safety monitoring dashboard** for underground tunnel construction. It provides **continuous environmental monitoring, bi-directional worker-admin communication, and automated danger alerts** — all in a single HTML file that runs in any browser.

 9 Live Sensors Temperature, CO, Methane, Smoke, Humidity, AQI, Vibration, CO2, Light	 23 Languages All 22 Scheduled Languages of India + Tibetan	 Two-Way Comms Workers report hazards; admins send instructions in real-time
 Auto Alerts Audio sirens + visual overlays triggered by threshold breaches	 12 Workers Simulated crew with roles, zones, gear checklists	 Zero Infrastructure Single HTML file, no backend, works offline

SYSTEM ARCHITECTURE

Single-File, Zero-Dependency Design

Layer	Component	Technology
Presentation	Dashboard UI, Charts, Worker Portal	HTML5, CSS3 Custom Properties
Logic	Sensor Engine, Alert System, i18n	Vanilla JavaScript (ES6+)
Data Layer	Sensor History, Reports, Contacts	In-Memory Arrays + localStorage
Simulation	Gaussian + Perlin noise sensor models	Box-Muller transform
Audio	Warning beeps, Danger sirens	Web Audio API (OscillatorNode)
Deployment	Global CDN, Auto-SSL	Vercel Static Hosting

■ ADMIN DASHBOARD

- 5×9 live sensor grid with sparklines
- Configurable danger thresholds
- Worker tracking per zone
- Report inbox with reply capability
- Emergency contacts panel
- Alert feed + audio warnings

■ WORKER PORTAL

- Native language interface (23 langs)
- Zone status with color coding
- Safety gear checklist per zone
- Hazard reporting with voice + photo
- Incoming admin instructions
- Real-time danger alerts

9 Environmental Parameters • 5 Tunnel Zones • Real-Time

Parameter	Unit	Range	Mean $\pm$ Std Dev	Danger Threshold
Temperature	°C	22–48	30.04 $\pm$ 5.02	38°C
Carbon Monoxide	ppm	0–92	19.96 $\pm$ 14.66	35 ppm
Methane	%LEL	0–13	2.91 $\pm$ 2.13	5% LEL
Smoke Level	$\mu\text{g}/\text{m}^3$	0–178	36.05 $\pm$ 27.19	70 $\mu\text{g}/\text{m}^3$
Humidity	%RH	27–96	64.89 $\pm$ 9.89	85% RH
Air Quality Index	—	0–473	102.84 $\pm$ 74.45	200
Vibration	g	0–0.15	0.05 $\pm$ 0.03	0.1 g
CO2 Level	ppm	0–1844	509.59 $\pm$ 279.89	1000 ppm
Light Intensity	lux	9–520	251.44 $\pm$ 79.2	100 lux

■ SAFE

All parameters normal

■■ WARNING

80–100% of threshold

■ DANGER

Threshold exceeded

TUNNEL ZONES

5 Operational Zones • Independent Monitoring • Zone-Specific Equipment

Zone	Name	Workers (base/max)	Required Equipment
A	Intake Shaft	4/8	Air quality monitor, Dehumidifier
B	Main Bore	8/15	Gas detector (CO/CH4), Cap lamp, Fire extinguisher
C	Deep Excavation	6/12	Multi-gas detector, Self-rescue device (SCSR), Rock bolt wrench
D	Tunnel Face	5/10	SCSR, Vibration sensor, Suppression system, High-power headlamp
E	Ventilation Bay	3/6	Anemometer, Wet-bulb thermometer

MANDATORY SAFETY GEAR (ALL ZONES)

- ✓ Hard Hat (Class E rated)
- ✓ Safety Goggles (Anti-fog, sealed)
- ✓ Respirator Mask (N95 / P100)
- ✓ Hi-Vis Vest (Class 3 reflective)

✓ Cut-Resistant Gloves

✓ Steel-Toe Boots (Slip-resistant)

✓ Ear Protection (NRR 25+ dB)

✓ Two-Way Radio (Ch. 7 tunnel
freq)

HAZARD REPORTING

Workers Report → Admin Responds → Loop Closes

Report Type	Description	Priority
■ Gas Smell	Unusual gas detected	■■ URGENT
■ Structural Damage	Crack or deformation spotted	■■ URGENT
■ Water Leak	Seepage or flooding	Standard
■ Unusual Noise	Rumbling or abnormal sound	Standard
■■ Excessive Heat	Temperature spike felt	■■ URGENT
■■ Equipment Failure	Malfunction detected	Standard
■■ Poor Visibility	Dust or smoke buildup	Standard
■ Electrical Hazard	Sparking or exposed wiring	■■ URGENT
■ Other	Any other abnormality	Standard



Worker

Files report with
voice + photo



Admin Inbox

Red alerts for urgent
Amber for standard



Response

Acknowledge + send
instructions back

23 LANGUAGES

Every Scheduled Language of India + Tibetan



Why it matters: India's tunnel workers are migrant laborers from every state. A safety system in English alone excludes the majority of its users. Ironhaven delivers every label, alert, and instruction in the worker's own script.

Built with Claude Opus 4.6 by Anthropic

Ironhaven was not built by a human programmer writing code line by line. It was created through **human–AI collaboration**, where the human defined the vision and the AI agent designed, coded, debugged, and iterated the entire application.



AI Capabilities Demonstrated

- **Full-Application Generation** — 3000+ lines of production-grade code from natural language
- **Domain Knowledge** — Tunnel safety protocols, hazard thresholds, construction roles
- **Iterative Debugging** — Diagnosed sandbox limitations, race conditions, DOM update conflicts
- **Cross-Cutting Design** — Accessibility, i18n, responsive layout, safety-critical UI patterns

CONSTRAINTS

Honest Limitations of the Current Prototype

- **Simulated Data** — All sensor readings are algorithmic; no real IoT sensors connected
- ■ **No Backend** — Data lives in browser memory + localStorage; no server-side persistence
- **Single-User** — Runs in one browser tab; no multi-device synchronization
- **No Authentication** — No login system or role-based access control
- **Browser Sandbox** — confirm(), tel: links, and alert() are restricted in hosted artifacts
- **Local-Only Comms** — Worker-admin messaging stays within the app; no external push
- **No GPS Tracking** — Worker position is zone-based simulation, not real indoor positioning

These are deliberate scope decisions, not failures. The prototype proves the concept; the next section shows how each constraint is resolved in production.

Production Architecture for Tunnel Management Organizations

Location	Infrastructure	Function
Tunnel Zone	Edge Gateway + Sensors	Local aggregation, 5s push interval
Tunnel Portal	On-Site Server Rack	Data processing, local DB, WebSocket server
Surface HQ	Cloud/On-Prem Cluster	Central dashboard, analytics, ML inference
Emergency	API Integration	Auto-alerts to fire/ambulance/rescue
Regulatory	Compliance Engine	Automated DGMS/OSHA safety logs

Recommended Sensor Hardware

Parameter	Suggested Sensor
Gas Detection	Honeywell BW Ultra / Drager X-am 8000
Temp/Humidity	Vaisala HMT330 (IP67, tunnel-rated)
Vibration	PCB Piezotronics 608A11 (triaxial)
Air Quality	TSI DustTrak / Aeroqual AQM 65

Parameter	Suggested Sensor
Smoke/Fire	Xtralis VESDA VLP (aspirating)

Backend Stack

- **Data Ingestion:** MQTT broker (Mosquitto/HiveMQ) from edge gateways
- **Time-Series DB:** InfluxDB or TimescaleDB with retention policies
- **App Server:** Node.js / FastAPI with real-time WebSocket feeds
- **ML Pipeline:** Anomaly detection (LSTM/Transformer) for predictive warnings
- **Auth:** RBAC — Admin, Supervisor, Worker roles

FUTURE ROADMAP

From Prototype to Production Platform

PHASE 1

3–6 months

- Real IoT sensor connection via MQTT
- User authentication + RBAC
- WebSocket real-time multi-device sync
- Progressive Web App (offline-first)

PHASE 2

6–12 months

- ML anomaly detection (LSTM/Transformer)
- Predictive alerts ("Methane trending to danger — ETA 12 min")
- Computer vision PPE compliance from CCTV
- NLP hazard report severity classification

PHASE 3

12–24 months

- Multi-tunnel regional command centers
- BIM integration for 3D tunnel visualization
- Regulatory automation (DGMS, ISO 45001)
- Wearable + smart helmet integration

PHASE 4

24+ months

- AI copilot for safety officers
- Digital twin with real-time 3D rendering
- Federated learning across tunnel sites
- Autonomous drone post-blast inspection

TECHNOLOGY STACK

Zero External Libraries • Pure Web Standards

Category	Technologies
Frontend	HTML5, CSS3 (Custom Properties, Flexbox, Grid), ES6+ JavaScript
Charting	Custom SVG sparkline renderer (zero library dependency)
Audio Alerts	Web Audio API (OscillatorNode + GainNode)
Media Capture	MediaRecorder API (voice), FileReader API (photos)
Persistence	localStorage (thresholds, contacts, language, theme)
Theming	CSS Custom Properties with dark/light toggle
Deployment	Vercel (static hosting, global CDN, auto-SSL)
AI Agent	Claude Opus 4.6 (Anthropic) via Claude Code
Documentation	ReportLab Platypus (Python PDF generation)

■ ZERO EXTERNAL JAVASCRIPT LIBRARIES

The entire application — charts, animations, audio, internationalization — is built with **native browser APIs only**. This eliminates supply-chain risk, reduces load time to near-instant, and ensures the app works even in bandwidth-constrained tunnel environments with spotty connectivity.

IRONHAVEN SAFETY INTELLIGENCE

*"Every tunnel deserves real-time awareness.
Every worker deserves safety in their own language.
Every administrator deserves data that saves lives."*

Built with ❤️ for worker safety
Anamika Tripathi & Claude (Anthropic)

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